

Working with Children in HCI: Research, Design and Evaluation

Janet C Read

University of Central Lancashire, Preston, UK

Shuli Gilutz

Tel Aviv University, Tel Aviv, Israel

ABSTRACT

Child Computer Interaction is concerned with the research, design, and evaluation of interactive technologies for children. Whilst many aspects of general HCI can be applied into this field, there are important adaptations to be made when conducting work for and with children throughout all stages of the design cycle. This course overviews the main tools and techniques in use by the CCI community presented alongside examples and experiences from academia and industry. The course is hands on and provides highly useful checklists and tips to ensure children (and researchers and developers) get the most out of participation in HCI activities.

CCS CONCEPTS

• **Human-centered computing** → Interaction design; Interaction design process and methods.

KEYWORDS

Child Computer Interaction, Evaluation, Design, Research, Ethics, Children, UX research

ACM Reference Format:

Janet C Read and Shuli Gilutz. 2021. Working with Children in HCI: Research, Design and Evaluation. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 3 pages. <https://doi.org/10.1145/3411763.3445004>

1 INTRODUCTION

Children around the world are consumers of, and users of, digital technology. The age at which they interact with technology is dropping with reports of babies as young as one interacting with tablet technologies [14]. At the other end of the spectrum, it is widely reported that teenagers spend lengthy times alone with computer technology, often playing online games or engaging in online chatter with friends they may or may not know [1]. More than ever before, we need to better understand how best to design these technologies and evaluate their instances of use so we can create digital worlds for children that are beneficial to them and that enhance their lives.

While children become more technology immersed, the gap in knowledge between adults who may be designing and mediating the use of such technologies, and the children using the technologies, is widening. As an example, a study in 2019 found that preschoolers aged 3 – 5 were 50% more able to use iPads than their teachers had

assumed [16]. In a recent (2020) study, Iivari et al. describe how, faced with online learning and a lack of physical social interaction, children have become masters of a whole raft of technologies [13]. A study funded by the BBC showed that whilst children are becoming expert, parents face many difficulties even in determining if applications are suitable for their children to use. These parents are looking to designers to better deal with issues of accessibility, of parental control and of quality content design [22].

Designers and academics, therefore, need effective ways to better understand children and their contexts. When conducting research with, or designing technology for, children and families, researchers must take into consideration unique factors, that are less relevant for the general population. As examples, research is needed to understand how children, given their limited vocabularies and their young voices, interact with new technologies like voice interfaces [7], designers need to make products intended for learning more accessible to children who may have motoric, sensory or cognitive challenges [5], and children need to be empowered to give their opinions of, and their ideas towards, new innovations [4].

The CHI community has been instrumental in driving forward research and design with and for children in its widest sense. There has been a set of highly influential papers including work on co-design [26], Interface design for children with special needs [11], and on tangibles [24]. Recent themes in CHI have included work with children with special needs [8], robots and children [15], and the application of voice assistants in homes [3]. These themes require robust research, effective design practices and sound evaluation protocols.

Bespoke courses on Child Computer Interaction are few. There are some offerings at master's level in a handful of Universities but most often, those working in this area come either from design or education or from a HCI background - often with little experience of child-centered HCI. Courses at venues like CHI therefore offer a unique opportunity for a set of individuals, often working solo or in small groups, without local expertise in CCI, to come together and learn.

The course presented here is designed to meet these pressing needs. Versions of it have been delivered at previous CHI conferences and other ACM venues and these have separately covered research, evaluation, and design [20]. [21]. [19]. This current version includes academic and industry research with children [9], and child-centered design. The course has been updated to address all three design stages (before, during, after) and to consider how we might ensure that the children we interact with have the best possible time whilst the researchers get the best possible input, combining both academic research with children & industry research with children.

2 BENEFITS

This course will introduce attendees to core values and key methods and techniques for work with children in CCI and Children's UX.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3445004>

Attendees will be given practical handouts and documentation including – planning sheets, toolkits, emergency activities and checklists. The course focuses on specialist elements of HCI and UX that work with and for children. At the end of the course it is expected that attendees will:

- Be able to locate and use techniques that will be useful in general Child-Centered Design especially where there is a limited opportunity to work directly with children.
- Better understand the unique characteristics of working with children in research, design, and evaluation, and be enabled to improve the experiences of the children they work with
- Have a clearer understanding of the impact of method choices on children's agency and engagement and on research data, evaluation results and design ideas.
- Gain knowledge of techniques for the ethical recruitment of children (including the D4CR guide), the managing of studies and the writing and reporting of work.
- Discover new methods and tools and feel empowered to apply them in their own work.

2.1 Intended Audience

This course is ideal for researchers, practitioners and designers who are interested in the research and design of digital environments for children (of all age groups). In past courses, participants included researchers, practitioners, and students.

2.2 Prerequisites

It is assumed that those attending have a basic understanding of HCI methods and/or UX research methods. Outside of that there are no prerequisites.

3 CONTENTS

The course is structured in three parts, designed for CHI2020's hybrid event. We have actively interspersed each session with three or more activities to ensure participant engagement in either online or face to face delivery.

Part 1: The uniqueness of children in HCI/UX. In this part we will review guiding principles on research and design for and with children, and will highlight some common pitfalls. We will look at expert evaluation using the SEEM protocol [2]. This part will include aspects of designing for children as it relates to guidelines. We will show how personas can be used for Child-Centered Design [10].

Part 2: Working with Children. This is the largest part of the course and it represents around 50% of the content. Taking the three headings of Research with Children, Design with Children and Evaluation with Children – this section will focus on methods used in participatory work. We will introduce a set of tools and methods that have been shown to be effective when working with children. Research Methods include the PETT toolkit (in press) which can be used to describe the expertise of a sample of children. We will use the management of Wizard of Oz experiments [12] to highlight some of the practicalities of working with children in research. Design Methods that are described include Co-Design and Layered elaboration [26], and Obstructed Theatre [17] and we will stress the importance of using ideas from children in appropriate

ways. Evaluation Methods include the MemoLine [25], This or That [27] and The Fun Toolkit [18] as well as discussion on planning evaluations and on carrying out observational and ethnographic work.

Part 3: Reporting and Ethics. The course closes with a discussion of reporting and ethics, and then gives ideas for where we can go to learn more. We will discuss the CHECK toolkits for ethical work [23] as well as the Designing for Children's Rights (D4CR) guidelines [6]. Practical advice on the recruitment of children and on reporting both to the scientific community, and also to the children, is included in this section of the course.

3.1 Practical Work

A series of exercises is interspersed with the course content as seen in the attached PowerPoint file. Attendees carry out evaluations, plan a design session, and critique research. They additionally complete an ethical checklist and a D4CR evaluation.

3.2 Resources

The course materials will be placed in a shared Dropbox folder, and also made available on the IDC website. Previous year's materials can be seen at <http://www.idc-sig.org/researchcourse>. Information about the instructors can be found on the ChiCI website (www.chici.org) and on Shuli Gilutz's website.

4 INSTRUCTOR BACKGROUND

Janet C Read is a Professor of Child Computer Interaction. Currently researching cross cultural CCI, childlike computing, the ethics of children's participation and visual programming, Prof. Read is the founding editor of the International Journal of Child Computer Interaction, a previous chair (on two occasions) of the ACM Interaction Design and Children (IDC) Conference and a former chair of the ACM Child Computer Interaction Community. She is widely published in CCI and HCI.

Dr Shuli Gilutz is a research fellow at Tel-Aviv university, and an instructor at the Shenkar Toy Invention Program. She is a Google for Startups accelerator UX research mentor, as well as a board member of the D4CR Association [6]. She has been researching the design of digital environments for children for over 20 years, in academia and industry, and has been an active member of the ACM IDC conference community. She consults companies on strategy, design and research for/with children, and is a global speaker on these topics.

REFERENCES

- [1] Alireza Atarodi, Marziyeh Rajabi and Ahmadrza Atarodi. 2020. Cell Phone Use and Social Alienation of Young Teenagers. *Library Philosophy and Practice* (2020), 1-10.
- [2] E Baauw, M. M Bekker and W Barendregt. 2005. A Structured Expert Evaluation Method for the Evaluation of Children's Computer Games. In *Interact 2005*. (2005), Rome, Italy. Springer, 457.
- [3] Erin Beneteau, Ashley Boone, Yuxing Wu, Julie A Kientz, Jason Yip and Alexis Hiniker. 2020. Parenting with Alexa: Exploring the Introduction of Smart Speakers on Family Dynamics. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. (2020), 1-13.
- [4] Louanne E Boyd, Kathryn E Ringland, Oliver L Haimson, Helen Fernandez, Maria Bistarkey and Gillian R Hayes. 2015. Evaluating a collaborative iPad game's impact on social relationships for children with autism spectrum disorder. *ACM Transactions on Accessible Computing (TACCESS)*, 7, 1 (2015), 1-18.

- [5] Suzanne Clarke and J McGhee. 2017. Understanding Accessible Gameplay at the Children's BBC. . In BHNCI. Sunderland. BCI,
- [6] D4CR 2020. Designing for Children's Rights.
- [7] Daniel Fitton, Janet C Read, Gavin Sim and Brendan Cassidy. 2018. Co-designing voice user interfaces with teenagers in the context of smart homes. In *Proceedings of the 17th ACM Conference on Interaction Design and Children*. (2018), ACM, 55-66.
- [8] Christopher Frauenberger, Katta Spiel, Laura Scheepmaker and Irene Posch. 2019. Nurturing constructive disagreement-Agonistic design with neurodiverse children. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. (2019), 1-11.
- [9] Shuli Gilutz 2019. Back to the Future: User Research as a Critical Tool for Novel Design. Emerald Publishing Limited,
- [10] Shuli Gilutz 2019. Child-Centered Design: Integrating Children's Rights and Ethics into the Heart of the Design Process. Routledge, 49-54.
- [11] Juan Pablo Hourcade, Stacy R Williams, Ellen A Miller, Kelsey E Huebner and Lucas J Liang. 2013. Evaluation of tablet apps to encourage social interaction in children with autism spectrum disorders. In *Proceedings of the SIGCHI conference on human factors in computing systems*. (2013), 3197-3206.
- [12] Johanna Hoysniemi and Janet Read. 2005. Wizard of Oz studies with children. In *Proceedings of Interact 2005 Workshop on Child Computer Interaction: Methodological Research*. (2005),
- [13] Netta Iivari, Sumita Sharma and Leena Ventä-Olkkonen. 2020. Digital transformation of everyday life—How COVID-19 pandemic transformed the basic education of the young generation and why information management research should care? *International Journal of Information Management* (2020), 102183.
- [14] Jackie Marsh, Jamal Lahmar, Lydia Plowman, Dylan Yamada-Rice, Julia Bishop and Fiona Scott. 2020. Under threes' play with tablets. *Journal of Early Childhood Research* (2020).
- [15] Oussama Metatla, Sandra Bardot, Clare Cullen, Marcos Serrano and Christophe Jouffrais. 2020. Robots for Inclusive Play: Co-designing an Educational Game With Visually Impaired and sighted Children. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. (2020), 1-13.
- [16] Daniel James Mourlam, Gabrielle A Strouse, Lisa A Newland and Haoyu Lin. 2019. Can they do it? A comparison of teacher candidates' beliefs and preschoolers' actual skills with digital technology and media. *Computers & Education*, 129 (2019), 82-91.
- [17] J. C Read, D Fitton and E Mazzone. 2010. Using Obstructed Theatre with Child Designers to Convey Requirements. In *CHI2010*. (2010), Atlanta, GA. ACM Press,
- [18] J. C Read and S. J MacFarlane. 2006. Using the Fun Toolkit and Other Survey Methods to Gather Opinions in Child Computer Interaction. In *Interaction Design and Children, IDC2006*. (2006), Tampere, Finland. ACM Press, 81 - 88.
- [19] J. C Read and P Markopoulos. 2008. Lifelong Interactions - Understanding children's interactions: evaluating children's interactive products. *Interactions*, 15, 6 (2008), 26 - 29.
- [20] Janet C Read. 2018. Doing research with children: a child computer interaction perspective. In *Proceedings of the 17th ACM Conference on Interaction Design and Children*. (2018), ACM, 745-747.
- [21] Janet C Read and Shuli Gilutz. 2016. Research Methods for Child Computer Interaction. In *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*. (2016), ACM, 927-930.
- [22] Janet C Read, Matthew Horton, Suzanne Clarke, Rhia Jones, Dan Fitton and Gavin Sim. 2018. Designing for the 'at home' experience of parents and children with tablet games. In *Proceedings of the 17th ACM Conference on Interaction Design and Children*. (2018), ACM, 441-448.
- [23] Janet C Read, Matthew Horton, Gavin Sim, Peggy Gregory, Daniel Fitton and Brendan Cassidy. 2013. CHECK: a tool to inform and encourage ethical practice in participatory design with children. In *CHI'13 Extended Abstracts on Human Factors in Computing Systems*. (2013), ACM, 187-192.
- [24] D Stanton, V Bayou, H Neale, A Ghuli, A Benford, S Cobb, Ingram. R. C O'Malley, J Wilson and T Pridmore. 2001. Classroom Collaboration in the Design of Tangible Interfaces for Storytelling. In *CHI 2001*. (2001), ACM Press, 482 - 489.
- [25] Jorick Vissers, Lode De Bot and Bieke Zaman. 2013. MemoLine: evaluating long-term UX with children. In *Proceedings of the 12th International Conference on Interaction Design and Children*. (2013), ACM, 285-288.
- [26] Greg Walsh, Alison Druin, Mona Leigh Guha, Elizabeth Foss, Evan Golub, Leshell Hatley, Elizabeth Bonsignore and Sonia Franckel. 2010. Layered elaboration: a new technique for co-design with children. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. Atlanta, Georgia, USA. ACM, 1237-1240. DOI:10.1145/1753326.1753512
- [27] B Zaman. 2009. Introducing a Pairwise Comparison Scale for UX Evaluations with Preschoolers. In *Interact 2009*. (2009), IFIP, 634 - 637.